

ECO387C - Fall 2024

Discussion 7

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Stochastic Endowment Economy

Consider identical individuals seeking to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}$$

where c_t denotes consumption and $0 < \beta < 1$ their discount factor. Agents operate in a perfectly competitive economic environment, where they trade a riskless one-period bond a_{t+1} that pays a gross interest rate R_t . The agents receive endowment y_t in each period. The endowment fluctuates between booms (y_h) and recessions (y_l), where $\infty > y_h > y_l > 0$. The two states change according to a two-state Markov chain where p is the probability of staying in the current state and $1 - p$ is the probability of switching to another state.

1. Write down the individual flow budget constraint.

$$c_t + a_{t+1} = R_t a_t + y_t$$

or

$$c_t + q_t a_{t+1} = a_t + y_t$$

where $q_t = R_{t+1}^{-1}$

2. What are the relevant state variables? a_t and y_t
3. Write the Bellman equation for this problem.

$$V(a, y) = \max_{a'} \left\{ \frac{(a + y - qa')^{1-\gamma}}{1-\gamma} + \beta \mathbb{E} [V(a', y') | y] \right\}$$

4. Derive the Euler equation.

- The first order condition with respect to a' :

$$-q(a + y - qa')^{-\gamma} + \beta \mathbb{E}[V_1(a', y') | y] = 0$$

where $V_1(a', y') = \frac{\partial V(a', y')}{\partial a'}$ denotes the derivative with respect to the first argument.

- To find $V_1(a', y')$, we first obtain $V_1(a, y)$. Suppose a'^* is the optimizer. Note that $a'^* = g(a, y)$. Then the value function is

$$V(a, y) = \frac{(a + y - qg(a, y))^{-\gamma}}{1 - \gamma} + \beta \mathbb{E}[V(g(a, y), y') | y]$$

Differentiating with respect to a and applying the chain rule:

$$\begin{aligned} V_1(a, y) &= (a + y - qa'^*)^{-\gamma} - q(a + y - qa'^*)^{-\gamma} g_1(a, y) + \beta \mathbb{E}[V_1(a'^*, y') | y] g_1(a, y) \\ &= (a + y - qa'^*)^{-\gamma} + \underbrace{\left(-q(a + y - qa'^*)^{-\gamma} + \beta \mathbb{E}[V_1(a'^*, y') | y] \right)}_{=0} g_1(a, y) \end{aligned}$$

Note that the underbraced term inside the bracket is equal to zero because of the first-order condition above (Envelope theorem). Then we have

$$V_1(a, y) = (a + y - qa')^{-\gamma} \Rightarrow V_1(a', y') = (a' + y' - q'a'')^{-\gamma}$$

- Combine the updated value function with the first-order condition:

$$q(a + y - qa')^{-\gamma} = \beta \mathbb{E}[(a' + y' - q'a'')^{-\gamma} | y]$$

This can be rewritten as

$$qc^{-\gamma} = \beta \mathbb{E}[c'^{-\gamma} | y] \Rightarrow q = \beta \mathbb{E} \left[\left(\frac{c'}{c} \right)^{-\gamma} \mid y \right]$$

5. Consider the equilibrium where the goods and asset markets clear. What is the equilibrium interest rate? Suppose $y_h = 2$, $y_l = 1$ and $\gamma = 1$, which corresponds to $u(c) = \log(c)$.

The market clearing condition requires $c_t = y_t$:

$$q_t = \mathbb{E}_t \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^{-1} \right] = \mathbb{E}_t \left[\beta \left(\frac{y_{t+1}}{y_t} \right)^{-1} \right]$$

Then, expanding the Euler equation:

- If $y_t = y_h$:

$$q_t = \mathbb{E}_t \left[\beta \left(\frac{y_{t+1}}{y_t} \right)^{-1} \right] = \beta \left(\frac{y_h}{y_h} \right)^{-1} p + \beta \left(\frac{y_l}{y_h} \right)^{-1} (1 - p) = \beta(2 - p)$$

- If $y_t = y_l$:

$$q_t = \mathbb{E}_t \left[\beta \left(\frac{y_{t+1}}{y_t} \right)^{-1} \right] = \beta \left(\frac{y_h}{y_l} \right)^{-1} (1 - p) + \beta \left(\frac{y_l}{y_l} \right)^{-1} p = \beta \left(\frac{1}{2} + \frac{1}{2} p \right)$$